2) $\int \frac{24x+12}{\sqrt{x^2+x+1}}\,dx$

$\sqrt{x^2+x+1}$

$\frac{d}{dx}(x^2+x+1)\ 2x+1$

$24x+12$

$24x+12=12(2x+1)$

$\int \frac{12(2x+1)}{\sqrt{x^2+x+1}}\,dx$

$du=(2x+1)\,dx$

$12\int \frac{1}{\sqrt{u}}\,du$

$\frac{1}{\sqrt{u}}=u^{-\frac{1}{2}} \qquad 12\int u^{-\frac{1}{2}}\,du$

$12\int u^{-\frac{1}{2}}\,du=12\cdot\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \qquad 12\cdot 2u^{\frac{1}{2}}$

$24u^{\frac{1}{2}}$

$u=x^2+x+1$

$24u^{\frac{1}{2}}=24\sqrt{x^2+x+1}$

$$\boxed{\int \frac{24x+12}{\sqrt{x^2+x+1}}\,dx=24\sqrt{x^2+x+1}+C}$$

VERIFICAMOS

$\frac{d}{dx}\left(24\sqrt{x^2+x+1}\right)$

$24\cdot\frac{2x+1}{2\sqrt{x^2+x+1}}$

$\frac{12(2x+1)}{\sqrt{x^2+x+1}}$ $\qquad \frac{24x+12}{\sqrt{x^2+x+1}}$ ✓